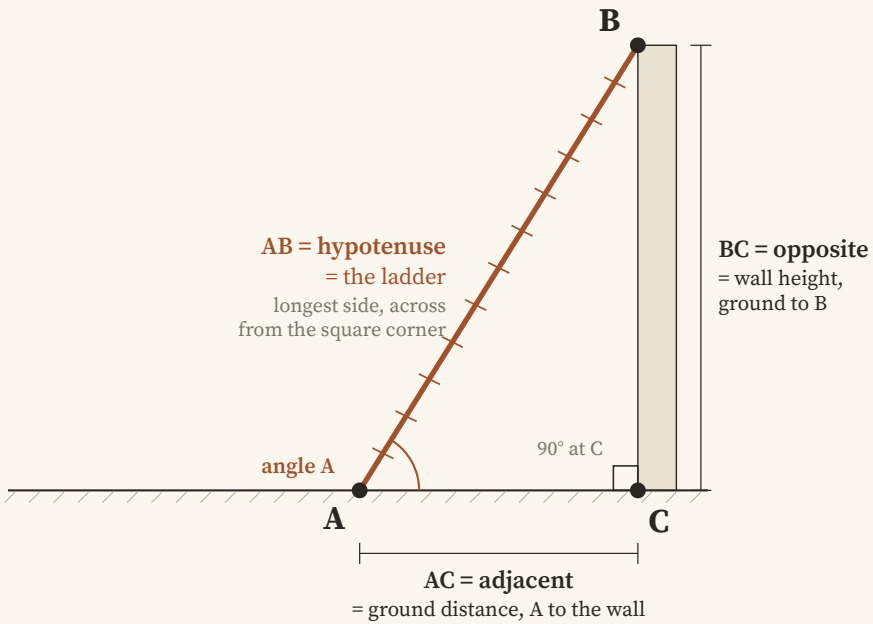
Sine, cosine and tangent: just a ladder against a wall

Every right triangle is a ladder, a wall and the ground. The three "trig" words are just three ways to divide its sides.

1 Name the three sides

The wall meets the ground in a square corner. The ladder closes the triangle. The names depend on which angle you look from: here, the angle at the foot of the ladder.

Three letters: **A** = foot of the ladder, on the ground. **B** = top of the wall, where the ladder touches it. **C** = bottom of the wall, the square corner. A side is named by its two end letters: AB is the ladder.



2 Three divisions, words first

"sin A" means the sine of the angle at A. Same for cos A and tan A.

sine written "sin"

How much of the ladder goes UP.

sin A = opposite ÷ hypotenuse = **BC ÷ AB**

cosine written "cos"

How much of the ladder goes ACROSS.

cos A = adjacent ÷ hypotenuse = **AC ÷ AB**

tangent written "tan"

Steepness: how much UP for each step ACROSS.

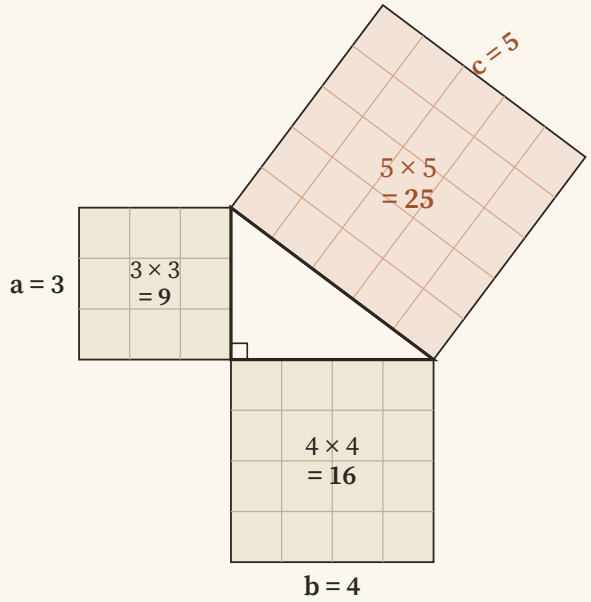
tan A = opposite ÷ adjacent = **BC ÷ AC**

Memory aid: SOH-CAH-TOA. Say it "so-ka-toe-a." Each three-letter chunk is one rule: **S**ine = **O**pposite ÷ **H**ypotenuse · **C**osine = **A**djacent ÷ **H**ypotenuse · **T**angent = **O**pposite ÷ **A**djacent.

Symbols on these pages ÷ divided by · × times · = equals · ≈ about equal · ° degrees of angle · "sin 30°" means "the sine of a 30° angle"

3 Pythagoras: count the squares

Take a triangle with sides 3, 4 and 5. Build a real square on each side and divide it into little unit squares.



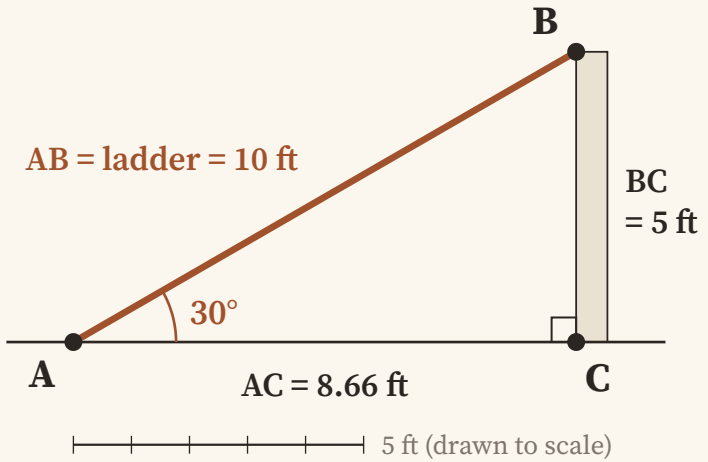
Count the little squares
on side a: 9
on side b: 16
together: 25
on side c: 25
The two small squares exactly fill the big one.

The short way to write it: $a^2 + b^2 = c^2$

a^2 (say "a squared") means **a × a**: the area of the square built on side a. So $3^2 = 3 \times 3 = 9$. The rule says: the two squares on the short sides add up to the square on the longest side. It works for every right triangle.

4 Worked example: a 10-ft ladder at 30°

The ladder is 10 ft long and makes a 30° angle with the ground. How high up the wall does it touch (B), and how far is its foot (A) from the wall (C)?



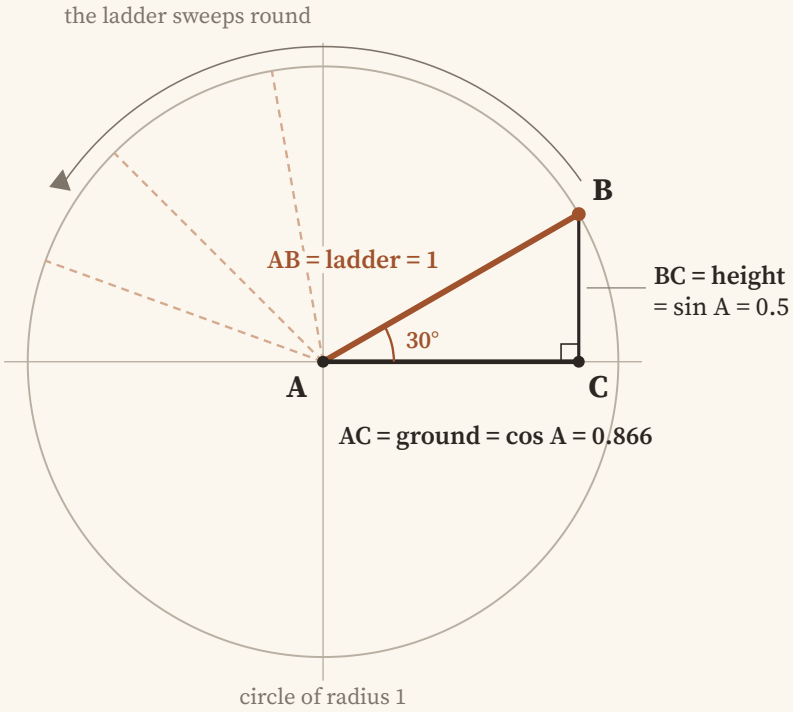
sin A = BC ÷ AB, so **BC = AB × sin A**. Likewise AC = AB × cos A. The values for 30° come from a calculator or table: sin 30° = 0.5, cos 30° ≈ 0.866.

Height	BC = 10 × sin 30° = 10 × 0.5 = **5 ft**
Ground	AC = 10 × cos 30° ≈ 10 × 0.866 ≈ **8.66 ft** (about 8.7)
Steepness	tan 30° = BC ÷ AC = 5 ÷ 8.66 ≈ **0.577** (0.577 ft up per foot across)
Check	$BC^2 + AC^2 = 5^2 + 8.66^2 = 25 + 75.0 \approx 100 = 10^2 = AB^2$ ✓

Use 8.66, not 8.7: $8.7^2 = 75.69$, which would give about 100.7.

5 Shrink the ladder to length 1

Make the ladder exactly 1 long. Dividing by 1 changes nothing, so **sine is simply the height** and **cosine is simply the ground distance**. Swing the ladder around its foot and its top traces a circle of radius 1 (radius: the distance from the center to the edge).



6 What the circle tells you

Same 30° as page 2. The 10-ft ladder reached 5 ft up and 8.66 ft across. The 1-long ladder reaches 0.5 up and 0.866 across: the same answers divided by 10.

Pythagoras again. Height, ground and ladder still make a right triangle, with ladder = 1:

$$BC^2 + AC^2 = AB^2 = 1^2$$

$$0.5^2 + 0.866^2 = 0.25 + 0.75 = 1$$

Written the short way: **$\sin^2 + \cos^2 = 1$**. ($\sin^2$ means sin × sin.) It is just the square-counting rule from page 2, on a ladder of length 1.

As the ladder swings up, the height (sine) grows from 0 to 1 and the ground distance (cosine) shrinks from 1 to 0.

GIS tie-in. This is why longitude lines squeeze together toward the poles: at 60° north, the circle around the Earth is cos 60° = 0.5 as long as the equator.

Symbols sin = sine · cos = cosine · tan = tangent · a^2 = a × a · ≈ about equal · ° degrees · Numbers rounded to 3 places.